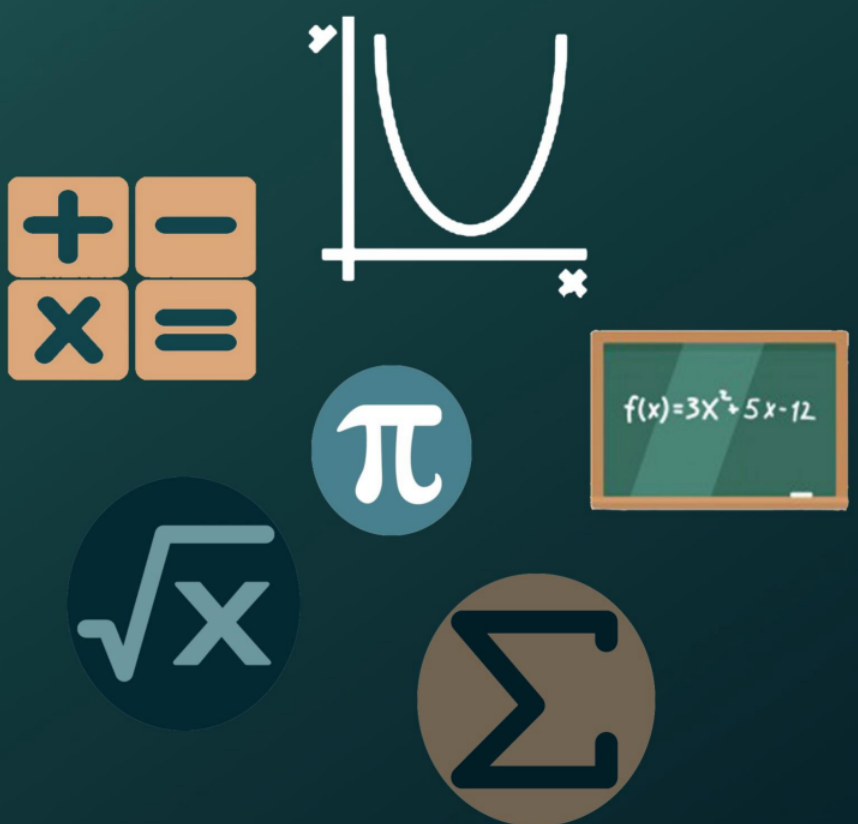
MATHEMATICS

Area Under the Curve
(Application of Integrals)



SHEET

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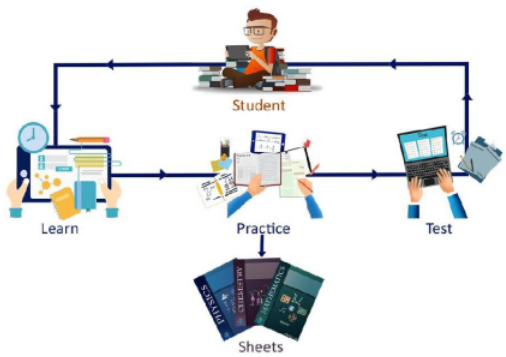
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

S. No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction to Area Under Curve	1-8	12		
2	Area Between Two Curves	9-15	1-4,7,8, 11,13,14	1-8	1-5,10-17, 21,23-27
3	Some Standard Areas	16	15-17	9	
4	Area Where the curve sketching is significant	17	5	10	22,28
5	Areas Dealing with Inequalities				6-9,18-20
6	Area by Shifting of Origin				
7	Area for Curves Represented Parametrically		18		
8	Area Enclosed by Inverse Function				
9	Determination of Unknown Parameters		6,9,10	11-13	30,31
10	Concept of Variable Area				
11	After Complete Chapter			14,15	29,32,33

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Area Under The Curve 1-1

Q1. The area between the curve $y = \sin^2 x$, x-axis and the ordinates $x = 0$ and $x = \frac{\pi}{2}$ is -

(A) π (B) $\pi / 2$
C) $\pi / 4$ (D) $\pi / 8$

Q2. The area bounded by the curve $y^2 = 4x$, y–axis and $y = 3$ is

(A) 2 sq. units (B) 9/4 sq. units
(C) 7/3 sq. units (D) 3 sq. units

Q3. Area bounded by curve $y = x^3$, x-axis and ordinates $x = 1$ and $x = 4$ is-

(A) 64 sq. units (B) 27 sq. units
(C) $\frac{127}{4}$ sq. units (D) $\frac{255}{4}$ sq. units

Q4. Area bounded by the parabola $y = 4x^2$, y-axis and the lines $y = 1$, $y = 4$ is-

(A) 3 sq. units (B) 7/5 sq. units
(C) 7/3 sq. units (D) None of these

Q5. The area bounded by the curve $y = \frac{1}{\cos^2 x}$, coordinates axes and $x = \pi/4$ is-

(A) 1 (B) 2
(C) $\pi/4$ (D) ∞

Q6. Area bounded by the lines $y = x$, $x = -1$, $x = 2$ and x-axis is-

(A) 5/2 sq. units (B) 3/2 sq. units
(C) 1/2 sq. units (D) None of these

Q7. The area bounded by the curve $y = x|x|$, x-axis and the ordinates $x = 1$, $x = -1$ is given by-

(A) 0 (B) $\frac{1}{3}$
(C) $\frac{2}{3}$ (D) None of these

Q8. The area of the region bounded by the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}\,(x, y > 0)$ and the coordinate axes is-

(A) $a^2/6$ (B) $a^2/2$
(C) $a^2/3$ (D) a^2

Area Between Two Curves 1-2

Q9. The area between the curve $x = 2y - y^2$ and y –axis is-

(A) 9/4 (B) 4/3
(C) 9 (D) None of these

Q10. The common area of the curves $y = \sqrt{x}$ and $x = \sqrt{y}$ is-

(A) 3 (B) 5/3
(C) 1/3 (D) None of these

Q11. The area between $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the straight line $\frac{x}{a} + \frac{y}{b} = 1$ is-

(A) $\frac{1}{2}ab$ (B) $\frac{1}{2}\pi ab$
(C) $\frac{1}{4}ab$ (D) $\frac{1}{4}\pi ab - \frac{1}{2}ab$

Q12. The area enclosed between the curve $y = \log_e(x+e)$ and the cooordinate axes is-

(A) 1 (B) 2 (C) 3 (D) 4

Q13. The area enclosed between the curves $y^2 = x$ and $x = |y|$ is -

(A) $\frac{2}{3}$ (B) 1 (C) $\frac{1}{6}$ (D) $\frac{1}{3}$

Q14. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to-

(A) $\frac{5}{3}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{4}{3}$

Q15. The area of the region enclosed by $y^2 = 8x$ and $y = 2x$ is

(A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Some Standard Areas 1-3

Q16. The whole area of the curve $9x^2 + 16y^2 = 144$ is

(A) 24π (B) 144π (C) 6π (D) 12π

Area by Shifting of Origin 1-6

Q17. The area bounded by $y = 4x - x^2$ and the x-axis is-

(A) 30/7 (B) 31/7 (C) 32/3 (D) 34/3

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If area bounded by the curves $y^2 = 4ax$ and $y = mx$ is $a^2/3$, then the value of m is-

(A) 2 (B) –2

(C) 1/2 (D) None of these

Q2. The value of area bounded by the curve $y = x^5$, $y = 1$ & $x = 0$ is greater than or equals to-

(A) $\frac{1}{2}$ (B) 1

(C) $\frac{6}{5}$ (D) $\frac{5}{6}$

Q3. Area bounded by curve $y = \cot^{-1}x$, $y = \tan^{-1}x$ and y-axis is equal to -

(A) $\int_0^{\pi/4} \tan x\,dx + \int_{\pi/4}^{\pi/2} \cot x\,dx$

(B) $\int_0^{\pi/4} \tan^{-1} x\,dx + \int_{\pi/4}^{\pi/2} \cot^{-1} x\,dx$

(C) $-\frac{2}{\pi}\int_0^{\pi/2} \log(\sin x)\,dx$

(D) $\int_{\pi/4}^{\pi/2} \tan^{-1} x\,dx + \int_0^{\pi/4} \cot^{-1} x\,dx$

Q4. Let $L : x - y - 1 = 0$ be a line and $C : y^2 = 2x + 1$ be a parabola, then

(A) area bounded by L and C lying above x-axis is $\frac{32}{3}$.

(B) area bounded by L and C is $\frac{16}{3}$.

(C) area bounded by L and C above x-axis is $\frac{9}{2}$.

(D) area bounded by L and C below x-axis is $\frac{5}{6}$.

Q5. Let $f(x) = 2 - |x - 1|$ & $g(x) = (x - 1)^2$, then-

(A) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{6}$

(B) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{3}$

(C) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{3}$

(D) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{6}$

Q6. Area enclosed by parabola $ay = 3(a^2 - x^2)$ and x-axis is 64, then value of 'a' is -

(A) 4 (B) 2

(C) –4 (D) –2

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques 14 to 15)

Let $f(x) = \begin{cases} x^3 & ; \ x \geq 0 \\ x^2 & ; \ x < 0 \end{cases}$ and $g(x) = \begin{cases} x^2 & ; \ x > 0 \\ x^3 & ; \ x \leq 0 \end{cases}$.

Q7. Area bounded by $y = f(x)$ and $y = g(x)$ is-

(A) $\frac{1}{5}$ (B) $\frac{1}{6}$

(C) $\frac{1}{11}$ (D) $\frac{1}{12}$

Q8. Area bounded by $f(x)$ and line $y = 1$ is-

(A) $\frac{17}{12}$ (B) $\frac{15}{6}$

(C) $\frac{3}{5}$ (D) $\frac{12}{11}$

Paragraph # 2 (Ques 9 to 10)

Let A be the area of region in the first quadrant bounded by the line $y = \frac{x}{2}$, the x-axis and the ellipse $\frac{x^2}{9} + y^2 = 1$. If m ($m > 0$) is such that A is equal to the area of the region in the first quadrant bounded by the line $y = mx$, the y-axis and the ellipse $\frac{x^2}{9} + y^2 = 1$, then :

Q9. Value of A is -

(A) $\frac{3}{2}\sin^{-1}\frac{2}{\sqrt{13}}$

(B) $3\sin^{-1}\frac{3}{\sqrt{13}}$

(C) $\frac{3}{2}\cos^{-1}\frac{2}{\sqrt{13}}$

(D) $3\cos^{-1}\frac{2}{\sqrt{13}}$

Q10. Value of m is -

(A) $\frac{2}{3}$ (B) $\frac{2}{9}$

(C) $\frac{2}{7}$ (D) $\frac{1}{9}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11. Let $f(x) = |x|$, $g(x) = |x - 1|$ and $h(x) = |x + 1|$.

Column-I	Column-II
(A) Area bounded by min (f(x), g(x)) and x-axis is	(P) $\frac{1}{8}$ sq. unit
(B) Area bounded by min (f(x), h(x)) and x-axis is	(Q) $\frac{1}{4}$ sq. unit
(C) Area bounded by min ((f(x), g(x), h(x)) and x-axis is	(R) $\frac{1}{2}$ sq. unit
(D) Area bounded by min (f(x), g(x), h(x)) and $y = \frac{1}{2}$ is	(S) $\frac{3}{4}$ sq. unit

SUBJECTIVE TYPE

Q12. Find the area bounded by the curve $y = mx$, x-axis and ordinates $x = 1$ and $x = 2$.

Q13. Find the area between the curves $y = x^3$ and $y = x$.

Q14. Find the area between the curves $y = 2x$, $x + y = 1$ and x-axis.

Q15. Find the whole area of circle $x^2 + y^2 = a^2$.

Q16. Find the area bounded by the parabola $y^2 = 4x$ and its latus rectum.

Q17. Find the area between the curve $y = \cos x$ and x-axis when $\pi/4 < x < \pi$.

Q18. Find the area bounded by the curves $x = a \cos t$, $y = b \sin t$ in the first quadrant.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If the ordinate $x = a$ divides the area bounded by the curve $y = \left(1 + \frac{8}{x^2}\right)$, x-axis and the ordinates $x = 2$, $x = 4$ into two equal parts, then $a =$

(A) 8 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$

Q2. The area of the small part bounded by the circle $x^2 + y^2 = 4$ and the line $x = 1$ is-

(A) $\frac{4\pi}{3} - \sqrt{3}$ (B) $\frac{2\pi}{3} - \sqrt{3}$

(C) $\frac{4\pi}{3} + \sqrt{3}$ (D) $\frac{5\pi}{3} + \sqrt{3}$

Q3. The area bounded by the circle $x^2 + y^2 = 1$ and the curve $|x| + |y| = 1$ is-

(A) $\pi - 2$

(B) $\pi - 2\sqrt{2}$

(C) $2(\pi - 2\sqrt{2})$

(D) none of these

Q4. Area bounded by the curve $y = \left[\frac{x^2}{64} + 2\right]$ (where [.] denotes the greatest integer function). $y = x - 1$ and $x = 0$ above the x-axis is-

(A) 2

(B) 3

(C) 4

(D) None of these

Q5. The area inside the parabola $5x^2 - y = 0$ but outside the parabola $2x^2 - y + 9 = 0$ is

(A) $8\sqrt{3}$

(B) $12\sqrt{3}$

(C) $4\sqrt{3}$

(D) none of these

Q6. The area of the region bounded by $y^2 = x$ and $x = 36$ is divided in the ratio 1 : 7 by the line $x = a$. then a equals-

(A) 7 (B) 8

(C) 9 (D) 0

Q7. The curve $f(x) = Ax^2 + Bx + C$ passes through the point (1, 3) and line $4x + y = 8$ is tangent to it at the point (2, 0). The area enclosed by $y = f(x)$, the tangent line and the y-axis is -

(A) 4/3 (B) 8/3

(C) 16/3 (D) 32/3

Q8. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is-

(A) 1 : 2 : 1

(B) 1 : 2 : 3

(C) 2 : 1 : 2

(D) 1 : 1 : 1

Q9. The area of the closed figure bounded by $y = x$, $y = -x$ & the tangent to the curve $y = \sqrt{x^2 - 5}$ at the point (3, 2) is -

(A) 5 (B) $2\sqrt{5}$

(C) 10 (D) $\frac{5}{2}$

Q10. The area bounded by the parabolas $y = (x + 1)^2$ and $y = (x - 1)^2$ and the line $y = 1/4$ is -

(A) 4 sq. units

(B) 1/6 sq. units

(C) 4/3 sq. units

(D) 1/3 sq. units

Q11. Let f(x) be a continuous function such that the area bounded by the curve $y = f(x)$, the x-axis and the two ordinates $x = 0$ and $x = a$ is $\frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a$. Then $f\left(\frac{\pi}{2}\right)$ is-

(A) $\frac{1}{2}$ (B) $\frac{\pi^2}{8} + \frac{\pi}{4}$

(C) $\frac{\pi + 1}{2}$ (D) $\frac{\pi}{2}$

Q12. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ell nx$ on the xy plane. Let D_1 denotes the region surrounded by C_1, C_2 and the line $x = 1$ and D_2 denotes the region surrounded by C_1, C_2 and the line $x = c$. If $D_1 = D_2$ then the value of 'c' -

(A) $\frac{e}{2}$ (B) e

(C) e – 1 (D) 2(e – 1)

Q13. If the area bounded by $y = ax^2$ and $x = ay^2$, $a > 0$, is 1, then a =

(A) 1 (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{3}$ (D) $\sqrt{3}$

Q14. Let f(x) be a non-negative continuous function such that the area bounded by the curve y= f(x), x-axis and the ordinates $x=\frac{\pi}{4}$ and $x=\beta>\frac{\pi}{4}$ is $\left(\beta\sin\beta+\frac{\pi}{4}\cos\beta+\sqrt{2}\beta\right)$. Then $f\left(\frac{\pi}{2}\right)$ is-

(A) $\left(\frac{\pi}{4}+\sqrt{2}-1\right)$

(B) $\left(\frac{\pi}{4}-\sqrt{2}+1\right)$

(C) $\left(1-\frac{\pi}{4}-\sqrt{2}\right)$

(D) $\left(1-\frac{\pi}{4}+\sqrt{2}\right)$

Q15. The area of the region between the curves $y=\sqrt{\frac{1+\sin x}{\cos x}}$ and $y=\sqrt{\frac{1-\sin x}{\cos x}}$ bounded by the lines x = 0 and $x=\frac{\pi}{4}$ is :-

(A) $\int_0^{\sqrt{2}-1}\frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

(B) $\int_0^{\sqrt{2}-1}\frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

(C) $\int_0^{\sqrt{2}+1}\frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

(D) $\int_0^{\sqrt{2}+1}\frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Area bounded by $|x-1|\le 2$ and $x^2-y^2=1$ is-

(A) $6\sqrt{2}+\frac{1}{2}\ln\left|3+2\sqrt{2}\right|$

(B) $6\sqrt{2}+\ln\left|3-2\sqrt{2}\right|$

(C) $6\sqrt{2}-\ln\left|3+2\sqrt{2}\right|$

(D) None of these

Q2. If A_n is the area bounded by $y=(1-x^2)^n$ and coordinate axes, $n\in N$, then

(A) $A_n=A_{n-1}$ (B) $A_n<A_{n-1}$

(C) $A_n>A_{n-1}$ (D) $A_n=2A_{n-1}$

Q3. Let 'a' be a positive constant number. Consider two curves $C_1: y=e^x$, $C_2: y=e^{a-x}$. Let S be the area of the part surrounding by C_1, C_2 and the y-axis, then -

(A) $\lim_{a\to-\infty}S=1$

(B) $\lim_{a\to0}\frac{S}{a^2}=\frac{1}{4}$

(C) Range of S is $[0,\infty)$

(D) S(a) is neither odd nor even

Q4. Area enclosed by the curve $y = \sin x$ between $x = 2n\pi$ to $x = 2(n+1)\pi$ is-

(A) $\int_0^{2\pi} \sin x\,dx$ (B) $2\int_0^{\pi} \sin x\,dx$

(C) $4\int_0^{\pi/2} \sin x\,dx$ (D) 4

Q5. For which of the following values of m, is the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals to 9/2?

(A) –4 (B) –2

(C) 2 (D) 4

Q6. The area enclosed between the curves $y^2 = x$ and $x^2 = y$ is equal to

(A) $2\int_0^1 \left(x - x^2\right) dx$

(B) 1/3

(C) area of the region $\left\{(x,y) : x^2 \le y \le |x|\right\}$

(D) None of these

Q7. Consider the following regions in the plane:
$R_1 = \{(x, y) : 0 \le x \le 1 \text{ and } 0 \le y \le 1\}$ and
$R_2 = \{(x, y) : x^2 + y^2 \le 4/3\}$
The area of the region $R_1 \cap R_2$ can be expressed as $\frac{a\sqrt{3} + b\pi}{9}$, where a and b are integers, then -

(A) a = 3 (B) a = 1

(C) b = 1 (D) b = 3

Q8. The area of the region of the plane bounded by $(|x|, |y|) \le 1$ & $xy \le \frac{1}{2}$ is -

(A) less then $4\ell n\ 3$ (B) $\frac{15}{4}$

(C) $2 + 2\ell n2$ (D) $3 + \ell n2$

Q9. Let A be the area bounded by the curve $f(x) = \frac{2}{1+x^4}$, x-axis y-axis and ordiante x = 1, then-

(A) $A > \frac{\pi}{2}$ (B) $A > 2\ell n2$

(C) $0 < A < \frac{\pi}{2}$ (D) $\frac{\pi}{2} < A < 2\ell n2$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques 10 to 14)

Let $f(x) = x^2 - 3x + 2$ be a function, $\forall\ x \in R$.

Q10. The area bounded by f(x), the x-axis and y-axis is-

(A) $\frac{1}{3}$ sq. unit (B) $\frac{2}{3}$ sq. unit

(C) $\frac{3}{5}$ sq. unit (D) $\frac{5}{6}$ sq. unit

Q11. The area bounded by the curve f(|x|) above x-axis is-

(A) $\frac{2}{5}$ sq. unit (B) $\frac{5}{2}$ sq. unit

(C) $\frac{3}{5}$ sq. unit (D) $\frac{5}{3}$ sq. unit

Q12. The area bounded by the curve |f(|x|)| between $1 \le |x| \le 2$ and x-axis is-

(A) $\frac{1}{5}$ sq. unit (B) $\frac{1}{4}$ sq. unit

(C) $\frac{1}{3}$ sq. unit (D) $\frac{1}{2}$ sq. unit

Q13. If $|f(|x|)| = a$ has 8 real roots, then-

(A) $a \in \left(0, \frac{1}{4}\right)$ (B) $a \in \left(\frac{1}{4}, 2\right)$

(C) $a \in \left(-\frac{1}{4}, 0\right)$ (D) None of these

Q14. The number of solution of $|y| = |f(|x|)|$ and $x^2 + y^2 = 2$ is-

(A) 4 (B) 6

(C) 8 (D) 5

Paragraph # 2 (Ques 15 to 17)

Consider the functions defined implicitly by the equation $y^3 - 3y + x = 0$ on various intervals in the real line. If $x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$. If $x \in (-2, 2)$, the equation implicitly defines a unique real valued differentiable function $y = g(x)$ satisfying $g(0) = 0$.

Q15. If $f(-10\sqrt{2}) = 2\sqrt{2}$, then $f''(-10\sqrt{2}) =$

(A) $\frac{4\sqrt{2}}{7^3 3^2}$ (B) $-\frac{4\sqrt{2}}{7^3 3^2}$

(C) $\frac{4\sqrt{2}}{7^3 3}$ (D) $-\frac{4\sqrt{2}}{7^3 3}$

Q16. The area of the region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is

(A) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + b f(b) - a f(a)$

(B) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + b f(b) - a f(a)$

(C) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - b f(b) + a f(a)$

(D) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - b f(b) + a f(a)$

Q17. $\int_{-1}^{1} g'(x)\, dx =$

(A) $2g(-1)$ (B) 0

(C) $-2g(1)$ (D) $2g(1)$

Paragraph # 3 (Ques 18 to 20)

Five curves defined as follows :

$C_1 : |x + y| \le 1$

$C_2 : |x - y| \le 1$

$C_3 : |x| \le \frac{1}{2}$

$C_4 : |y| \le \frac{1}{2}$

$C_5 : 3x^2 + 3y^2 = 1$

Q18. The area bounded by C_1 and C_2 which does not contain the area bounded by C_5, is -

(A) $2 - \frac{\pi}{4}$ (B) $2 - \frac{\pi}{6}$

(C) $2 - \frac{\pi}{3}$ (D) 2

Q19. That part of area of curve C_5 which does not contain points satisfying C_3 and C_4, is-

(A) $\frac{\pi}{3} - \frac{1}{2}$ (B) $\frac{\pi}{3} - 1$

(C) $\frac{\pi}{3} - \frac{1}{6}$ (D) $\frac{2\pi}{9} - \frac{1}{\sqrt{3}}$

Q20. That part of area which is bounded by C_1 and C_2 but not bounded by C_3 and C_4, is -

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) none of these

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q21.

Column-I	**Column-II**
(A) The cosine of the angle between the curves $y = 3^{x-1} \ln x$ and $y = x^x - 1$ at their point of intersection on the line $y = 0$, is	(P) 0
(B) The area bounded by the curves $x = -4y^2$ and $(x - 1) = -5y^2$ is	(Q) 1
(C) The value of the integral $\int_0^{\pi/2} (\sin x)^{\cos x} (\cos x \cot x - \ln(\sin x)^{\sin x})\, dx$, is	(R) $\frac{4}{3}$
(D) A continuous function $f : [1, 6] \to [0, \infty)$ is such that $f'(x) = \frac{2}{x + f(x)}$ and $f(1) = 0$, then the maximum value of f cannot exceed.	(S) $2 \ln 6$

Q.22

Column - I	**Column-II**
(A) The area bounded y the curve $y = x \lvert x \rvert$, x-axis ordinates $x = 1, x = -1$	(P) $\frac{10}{3}$ sq. units
(B) The area of the region lying between the line $x - y + 2 = 0$ and the curve $x = \sqrt{y}$	(Q) $\frac{64}{3}$ sq. units
(C) The area enclosed between the curves $y^2 = x$ and $y = \lvert x \rvert$	(R) $\frac{2}{3}$ sq. units
(D) The area bounded by parabola $y^2 = x$, straight line $y = 4$ and y-axis	(S) $\frac{1}{6}$ sq. units

SUBJECTIVE TYPE

Q23. Find the area bounded by the curve $xy^2 = a^2 (a - x)$ and x-axis

Q24. Find the area between the curve $y = x (x - 1) (x - 2)$ and x-axis.

Q25. Find the area bounded by the curves $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$.

Q26. Find the value(s) of the parameter 'a' ($a > 0$) for each of which the area of the figure bounded by the straight line, $y = \frac{a^2 - ax}{1 + a^4}$ & the parabola $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ is the greatest.

Q27. If $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2 + 3a \\ 3b^2 + 3b \\ 3c^2 + 3c \end{bmatrix}$,

f(x) is a quadratic function and its maximum value occurs at a point V. A is a point of intersection of y = f(x) with x-axis and point B is such that chord AB subtends a right angle at V. Find the area enclosed by f(x) and chord AB.

Q28. The area bounded by curves $3x^2 + 5y = 32$ and $y = |x - 2|$ is-

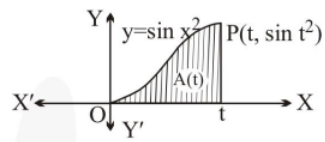
Q29. The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

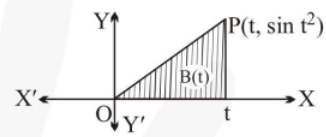
Q30. Find the values of m ($m > 0$) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is, (i) 9/2 square units & (ii) minimum. Also find the minimum area.

Q31. Consider the curve $y = x^n$ where $n > 1$ in the 1st quadrant. If the area bounded by the curve, the x-axis and the tangent line to the graph of $y = x^n$ at the point (1, 1) is maximum then find the value of n.

Q32. Let $f : [0, \infty) \to R$ be a continuous and strictly increasing function such that $f^3(x) = \int_0^x t f^2(t)\,dt$, $\forall\ x > 0$. Find the area enclosed by $y = f(x)$, the x-axis and the ordinate at $x = 3$.

Q33. The figure shows two regions in the first quadrant.





A(t) is the area under the curve $y = \sin x^2$ from 0 to t and B(t) is the area of the triangle with vertices O, P and M(t, 0). Find $\lim_{t\to 0} \frac{A(t)}{B(t)}$.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	B	D	C	A	A	C	A	B	C
11	**12**	**13**	**14**	**15**	**16**	**17**			
D	A	C	D	A	D	C			

Ex - 1A

1. A,B **2.** A,D **3.** A,C **4.** B,C,D **5.** B,C

6. A,C **7.** D **8.** A **9.** C **10.** B)

11. (A) → (Q); (B) → (Q); (C) → (R); (D) → (S) **12.** $\left(\frac{3}{2}\right)m$ **13.** $\frac{1}{2}$

14. $\frac{1}{3}$ **15.** πa^2 **16.** $\frac{8}{3}$ **17.** $\frac{2\sqrt{2}-1}{\sqrt{2}}$ **18.** $\frac{\pi ab}{4}$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	A	A	C	B	C	B	D	A	D
11	**12**	**13**	**14**	**15**					
A	B	B	D	B					

EX - 2A

1. C, B **2.** B **3.** A,B,C,D **4.** B,C,D **5.** B,D

6. A, B, C **7.** A,C **8.** A,D **9.** A,B **10.** D

11. D **12.** C **13.** A **14.** C **15.** B

16. A **17.** D **18.** C **19.** D **20.** A

21. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (S)

22. (A) $\rightarrow$ (R); (B) $\rightarrow$ (P); (C) $\rightarrow$ (S); (D) $\rightarrow$ (Q)

23. πa^2 **24.** $\frac{1}{2}$ **25.** $\frac{1}{3}$ sq. units **26.** $a = 3^{1/4}$ **27.** $\frac{125}{3}$ sq. units.

28. $\frac{33}{2}$ **29.** $\frac{3\pi+2}{\pi-2}$ **30. (i)** $m = 1$, **(ii)** $m = \infty$; $A_{min} = 4/3$ **31.** $\sqrt{2}+1$

32. 3/2 **33.** 2/3

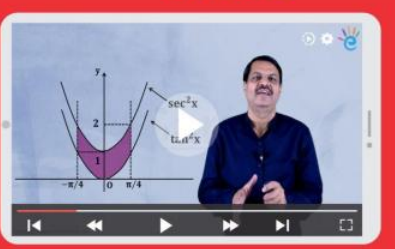
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